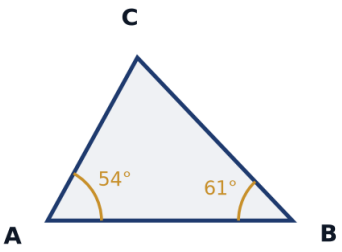
Proving lines parallel

Core Concepts: Proving lines parallel

1 In $\triangle ABC$ studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 92°

b. 102°

c. 84°

d. 137°

2 A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 14

c. 12

d. 15

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 85°

b. 95°

c. 77°

d. 137°

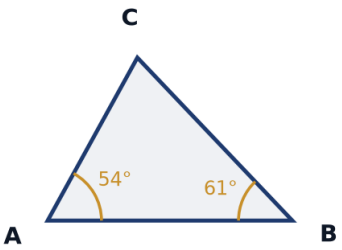
5 A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm

c. 28 cm

d. 63 cm



6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 12

b. 17

c. 15

d. 18

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 78°

b. 88°

c. 70°

d. 137°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm

c. 35 cm

d. 112 cm

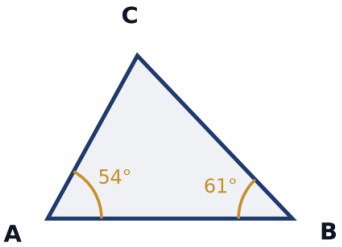
9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$?

a. 15

b. 20

c. 18

d. 21



Exam-Style Problem Solving & Modeling

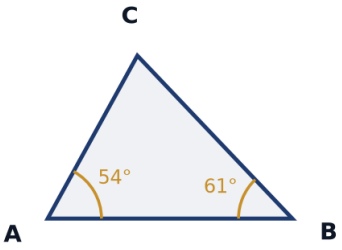
10 In $\triangle ABC$ studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 71°

b. 81°

c. 63°

d. 137°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm

c. 42 cm

d. 175 cm

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$?

a. 18

b. 23

c. 21

d. 24

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